

- 5 A freshwater fish species, *Oryzias latipes*, has individuals with four body colour patterns, as shown in Table 5.1.

Table 5.1

phenotype	body colour pattern
red	
white	
red with black spots	
white with black spots	

Two unlinked genes determine the body colour patterns shown in Table 5.1.

One gene controls whether the body colour is red or white:

- dominant allele **R** = red
- recessive allele **r** = white.

The other gene controls whether black spots are present or **not** present:

- dominant allele **B** = with black spots
- recessive allele **b** = without black spots.

A fish that is homozygous recessive at both loci is white.

Genetic crosses were carried out to investigate the inheritance of the four different body colour patterns.

Males that were red with black spots, and homozygous at both loci, were crossed with females that were white. The F1 offspring were all red with black spots.

These F1 offspring were then crossed to produce the F2 generation.

- (a) Table 5.2 shows the observed numbers obtained of each of the four different phenotypes for the F2 generation.

Table 5.2

phenotype	observed	expected	$O-E$	$(O-E)^2$	$\frac{(O-E)^2}{E}$
red with black spots	279	281.25			
white with black spots	95	93.75	1.25	1.5625	0.017
red	96	93.75	2.25	5.0625	0.054
white	30	31.25			
				$\chi^2 = \dots\dots\dots$	

Table 5.2 compares the observed numbers with the numbers that would be expected in the F2 generation for a normal dihybrid ratio.

Calculate χ^2 for the F2 generation by completing Table 5.2.

The formula for χ^2 is:

$$\chi^2 = \sum \frac{(O-E)^2}{E} \quad [3]$$

- (b) The critical value at $p = 0.05$ and 3 degrees of freedom is 7.815.

Comment on whether the null hypothesis should be accepted or rejected.

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Further analysis of the results from the F2 generation in Table 5.2 showed that there were no white males or white males with black spots.

In *O. latipes*, females have two **X** chromosomes and males have an **X** and a **Y** chromosome.

It was deduced that, in *O. latipes*:

- the gene that controls body colour is located on the **X** chromosome **and** the **Y** chromosome
- the gene that controls whether black spots are present or **not** is located on an autosome.

- (c) To produce the F₂ generation, red males with black spots, $X^R Y^R Bb$, were crossed with red females with black spots, $X^R X^r Bb$.

Complete the Punnett square in Fig. 5.1 to show the genotypes and phenotypes of the F₂ generation.

- Use the symbols X^R , X^r and Y^R for the alleles of the gene that controls body colour.
- Use the symbols B and b for the alleles of the gene that controls whether black spots are present or **not**.

Some of Fig. 5.1 has been completed for you.

male gametes	female gametes			
	$X^R B$	$X^r B$	$X^R b$	$X^r b$
$X^r B$	$X^R X^r BB$		$X^R X^r Bb$	$X^r X^r Bb$
	female		female	female
	red + black spots		red + black spots	white + black spots
$Y^R B$	$X^R Y^R BB$	$X^r Y^R BB$	$X^R Y^R Bb$	$X^r Y^R Bb$
	male	male	male	male
	red + black spots	red + black spots	red + black spots	red + black spots
$X^r b$	$X^R X^r Bb$	$X^r X^r Bb$	$X^R X^r bb$	
	female	female	female	
	red + black spots	white + black spots	red + no spots	
$Y^R b$		$X^r Y^R Bb$		$X^r Y^R bb$
		male		male
		red + black spots		red + no spots

Fig. 5.1

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- (d) Explain why there are no white males or males that are white with black spots in the F₂ generation.

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- (e) In another cross, red males with the genotype $X^R Y^R bb$ were mated with white females with the genotype $X^r X^r bb$. All the male offspring were expected to be red and all the female offspring were expected to be white.

The observed results showed that the offspring included two red females out of 253 and one white male out of 198.

Suggest an explanation for this unexpected result.

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